

Updated Partial Results and Failed Full-Push Analysis for the Smooth Non-Complete Norm Question

Automatic Math Discovery Run fa_banach_001

June 19, 2026

Status

This packet is a corrected bookkeeping and partial-result report for the question in Azagra–Dobrowolski–Garcia-Bravo [1]. It should not be read as a full solution. The exact C^1 question remains open in the literature searched for this run.

The earlier active packet proved correct partial statements, but its novelty claim for the $c_0(\Gamma)$ -injectable case was too strong. A supplied audit points out that this case is already contained in Dobrowolski’s 1979 paper [4, Proposition 5.1]: his condition (L) is precisely the existence of a continuous linear injection into some $c_0(\Lambda)$, and he proves existence of a non-complete C^∞ -smooth norm under that condition. Thus Theorem 2 below is retained as a useful alternative proof and as a bridge to later reductions, not as a new theorem.

The genuinely useful output of the present investigation is the following:

- a verified self-contained proof of the separable case;
- a verified self-contained alternative proof of the known $c_0(\Gamma)$ -injectable / Markuševič-basis case;
- a quotient-extension partial theorem;
- a likely valid reflexive-subspace partial theorem, whose infimal convolution smoothness step is the main item for human review;
- a record of why several attempted full-proof routes fail.

Source Question

The source paper asks, on page 13 immediately before Lemma 2.10:

It is not known whether every infinite-dimensional Banach space with a C^1 equivalent norm possesses a C^1 smooth non-complete norm.

The same passage notes that for C^k , $k \geq 2$, the corresponding answer is positive by D’Alessandro–Hajek [2].

It is not known whether every infinite-dimensional Banach space with a C^1 equivalent norm possesses a C^1 smooth non-complete norm.¹ The following lemma shows that for every Banach space with a C^p smooth norm there exists a kind of C^p *asymmetric non-complete subadditive functional* which successfully replaces the smooth non-complete norm in Bessaga's technique [14] for extracting points.

Lemma 2.10. *Let $(E_2, \|\cdot\|)$ be an infinite-dimensional Banach space which admits a (not necessarily equivalent) C^p smooth norm, where $p \in \mathbb{N} \cup \{\infty\}$. Then there exists a continuous function $\omega : E_2 \rightarrow [0, \infty)$ which is C^p smooth on $E_2 \setminus \{0\}$ and satisfies the following properties:*

- (1) $\omega(x + y) \leq \omega(x) + \omega(y)$, and, consequently, $\omega(x) - \omega(y) \leq \omega(x - y)$, for every $x, y \in E_2$;
- (2) $\omega(rx) = r\omega(x)$ for every $x \in E_2$, and $r \geq 0$;
- (3) $\omega(x) = 0$ if and only if $x = 0$;

Known and Verified Positive Cases

All Banach spaces are real. A norm p on X is continuous if it is continuous for the original Banach norm, and is non-complete if (X, p) is not complete.

Theorem 1 (Separable case). *Let X be an infinite-dimensional separable Banach space. Then X admits a continuous C^∞ -smooth non-complete norm.*

Proof. Choose a sequence (u_n) dense in the unit sphere of X . For each n , pick $x_n^* \in S_{X^*}$ with $x_n^*(u_n) = 1$. The family (x_n^*) separates points. Indeed, if $x \neq 0$, choose n such that $\|u_n - x/\|x\|\| < 1/2$. Then

$$x_n^*(x/\|x\|) \geq x_n^*(u_n) - \|u_n - x/\|x\|\| > 1/2.$$

Define

$$T : X \rightarrow \ell_2, \quad Tx = (2^{-n}x_n^*(x))_{n=1}^\infty.$$

Then T is bounded, injective, and compact, since its finite-rank truncations converge to T in operator norm. Set

$$p(x) = \|Tx\|_{\ell_2}.$$

This is a continuous norm on X , and it is C^∞ -smooth on $X \setminus \{0\}$ because the Hilbert norm is C^∞ -smooth away from zero and T is injective.

If (X, p) were complete, then T would identify (X, p) isometrically with a closed subspace of ℓ_2 . The open mapping theorem would make T , regarded as an operator from the original Banach space X , bounded below. A compact operator bounded below on an infinite-dimensional Banach space is impossible. Hence p is not complete. $\square$

Lemma 1 (Diagonal damping in $c_0(\Gamma)$). *Let Y be an infinite-dimensional linear subspace of $c_0(\Gamma)$. Then there is a bounded injective diagonal operator*

$$D : c_0(\Gamma) \rightarrow c_0(\Gamma), \quad (Da)(\gamma) = w_\gamma a(\gamma),$$

with $0 < w_\gamma \leq 1$, such that $D|_Y$ is not bounded below.

Proof. We inductively construct $y_n \in Y$ and pairwise disjoint finite sets $F_n \subseteq \Gamma$. Put $H_0 = \emptyset$. Having chosen $H_{n-1} = F_1 \cup \dots \cup F_{n-1}$, the space

$$Y \cap \bigcap_{\gamma \in H_{n-1}} \ker \delta_\gamma$$

is still infinite-dimensional, because only finitely many coordinate conditions have been imposed. Choose y_n in this space with $\|y_n\|_\infty = 1$. Since $y_n \in c_0(\Gamma)$, choose a finite set $E_n \subseteq \Gamma$ such that

$$|y_n(\gamma)| \leq 2^{-n} \quad (\gamma \notin E_n).$$

Set $F_n = E_n \setminus H_{n-1}$. Then y_n vanishes on the old coordinates and satisfies

$$|y_n(\gamma)| \leq 2^{-n} \quad (\gamma \notin F_n).$$

Define $w_\gamma = 2^{-n}$ for $\gamma \in F_n$, and $w_\gamma = 1$ outside $\bigcup_n F_n$. The associated diagonal operator D is bounded and injective. For every n , the coordinates in F_n are damped by 2^{-n} , while all coordinates outside F_n already have modulus at most 2^{-n} . Hence $\|Dy_n\|_\infty \leq 2^{-n}$, proving that $D|_Y$ is not bounded below. $\square$

Theorem 2 (Known $c_0(\Gamma)$ -injectable case). *Let X be an infinite-dimensional Banach space. Suppose that there is a bounded injective linear operator $J : X \rightarrow c_0(\Gamma)$ for some set Γ . Then X admits a continuous C^∞ -smooth non-complete norm.*

Proof. Let $Y = JX$. By Lemma 1, choose a bounded injective diagonal operator $D : c_0(\Gamma) \rightarrow c_0(\Gamma)$ such that $D|_Y$ is not bounded below. Then $DJ : X \rightarrow c_0(\Gamma)$ is injective and not bounded below.

Let ρ be a C^∞ -smooth equivalent norm on $c_0(\Gamma)$. Define $p(x) = \rho(DJx)$. Since DJ is injective, p is a norm; since ρ is smooth away from the origin, p is C^∞ -smooth on $X \setminus \{0\}$. If (X, p) were complete, the open mapping theorem would imply that p is equivalent to the original norm. Since ρ is equivalent to $\|\cdot\|_\infty$, this would make DJ bounded below, contradicting the construction. $\square$

Remark 1 (Novelty correction). Theorem 2 is not claimed as new. The supplied audit identifies it with Dobrowolski's condition (L) in [4, Proposition 5.1]. It is kept here because the short diagonal proof is useful for review and for the subsequent reductions.

Corollary 1 (Markuševič bases). *Every infinite-dimensional Banach space with a Markuševič basis admits a continuous C^∞ -smooth non-complete norm.*

Proof. Let $(x_\gamma, x_\gamma^*)_{\gamma \in \Gamma}$ be an M-basis. After normalizing the coordinate functionals, $f_\gamma = x_\gamma^* / \|x_\gamma^*\|$, the map

$$x \longmapsto (f_\gamma(x))_{\gamma \in \Gamma}$$

is a bounded linear injection into $c_0(\Gamma)$. The c_0 -condition follows by approximating x by finite linear combinations of the basis vectors. Theorem 2 applies. $\square$

A Further Quotient-Extension Partial Result

Proposition 1. *Let X be an infinite-dimensional Banach space. Suppose there is a closed, infinite-dimensional separable subspace $Y \subset X$ such that X/Y admits a C^1 -smooth equivalent norm. Then X admits a continuous C^1 -smooth non-complete norm.*

Proof. Choose a sequence $(y_n^*) \subset S_{Y^*}$ separating the points of Y , and extend each y_n^* by Hahn–Banach to $x_n^* \in S_{X^*}$. Let

$$T : X \rightarrow \ell_2, \quad Tx = (2^{-n} x_n^*(x))_{n=1}^\infty.$$

Then T is compact and $T|_Y$ is injective. Let $Q : X \rightarrow X/Y$ be the quotient map, and put

$$S : X \rightarrow (X/Y) \oplus_2 \ell_2, \quad Sx = (Qx, Tx).$$

If $Sx = 0$, then $x \in Y$ and $Tx = 0$, hence $x = 0$. Thus S is injective. Because $T|_Y$ is compact on an infinite-dimensional space, it is not bounded below, so S is not bounded below.

Let q be a C^1 -smooth equivalent norm on X/Y , and define

$$\rho(z, u) = (q(z)^2 + \|u\|_2^2)^{1/2}.$$

This is a C^1 -smooth equivalent norm on $(X/Y) \oplus_2 \ell_2$. The pullback $p(x) = \rho(Sx)$ is a continuous C^1 -smooth norm on X . If it were complete, the open mapping theorem would make S bounded below, a contradiction. $\square$

Corollary 2. *The source question has a positive answer whenever X contains a complemented infinite-dimensional separable subspace.*

Proof. If $X = Y \oplus Z$ with Y separable and infinite-dimensional, then $X/Y \cong Z$. The restriction of the given C^1 -smooth equivalent norm to Z gives the required smooth norm on the quotient. $\square$

Reflexive-Subspace Extension: Likely Valid, Review Carefully

The strongest additional positive route from the extended audit is the following. It is included as a human-review target rather than as the headline theorem of this packet, because the proof uses a smoothness assertion for an infimal-convolution norm.

Proposition 2 (Reflexive subspace route). *Let X admit a C^1 -smooth equivalent norm. If X contains an infinite-dimensional reflexive subspace, then X admits a continuous C^1 -smooth non-complete norm.*

Proof sketch for review. Choose a separable infinite-dimensional reflexive subspace $Y \subset X$. By Theorem 1, Y has a continuous smooth non-complete norm s . Let r be the given C^1 -smooth equivalent norm on X , and define

$$p(x) = \inf_{y \in Y} (r(x - y)^2 + s(y)^2)^{1/2}.$$

This is a norm. Positivity follows because if $p(x) = 0$, then a minimizing sequence forces $x \in Y$ and $s(x) = 0$. Reflexivity of Y gives minimizers for the infimum: bounded minimizing sequences have weakly convergent subsequences, and both r and s are weakly lower semicontinuous.

The intended smoothness proof uses the dual formula for this infimal convolution and the Smulyan lemma: the dual unit ball is described by the quadratic combination of r^* and the dual norm induced by s on restrictions to Y . Smoothness of r and s gives uniqueness and continuous dependence of supporting functionals. This is the step that should be checked by a human expert.

Finally, $p|_Y \leq s$. Since s is non-complete on Y , there are vectors in Y witnessing failure of bounded-below equivalence; hence p is not equivalent to the original norm on X . If (X, p) were complete, the open mapping theorem would force equivalence with the original norm, contradiction. $\square$

A Universal Non-Complete Norm and the Remaining Smoothing Problem

Every infinite-dimensional Banach space admits a continuous non-complete norm. Indeed, let (e_n) be a normalized basic sequence, let $Y = [e_n]$, let $Q : X \rightarrow X/Y$ be the quotient map, and extend

the coordinate functionals of Y to X^* . Then

$$p_0(x) = \|Qx\| + \sum_{n=1}^{\infty} 2^{-n} |e_n^*(x)|$$

is a continuous norm and satisfies $p_0(e_n) = 2^{-n}$. Thus the full source question can be viewed as a smoothing problem: under the hypothesis of a C^1 -smooth equivalent norm, can one smooth such a weaker non-complete norm without losing the long directions of its unit ball?

The finite-stage dual regularization route considers

$$K_m = \left\{ f \in X^* : r^*(f)^2 + \sum_{n=1}^m c_n |f(e_n)|^2 \leq 1 \right\}, \quad p_m = h_{K_m}.$$

Each finite-stage norm p_m has a controlled smoothness argument, and $p_m \downarrow p$ with $p(e_n) \leq c_n^{-1/2} \rightarrow 0$. The missing statement is norm convergence, locally uniformly in x , of the unique maximizers for $p_m(x)$. Weak-star convergence alone is not enough for C^1 smoothness of the limit.

Failed Full-Proof Routes

Symmetrizing the source paper's asymmetric functional

The source paper constructs a smooth asymmetric subadditive functional ω with one-sided non-completeness. The tempting norm

$$x \mapsto \omega(x) + \omega(-x)$$

is symmetric and smooth, but the source estimates only control one side. The sequence for which $\sum \omega(y_k) < \infty$ need not satisfy any corresponding estimate for $\omega(-y_k)$. This route does not prove non-completeness.

Trying to manufacture an M-basis from smoothness

A C^1 -smooth norm gives unique support functionals at nonzero points, but it does not make arbitrary growing closed spans proximal. The transfinite attempt to build an M-basis stalls exactly at the need for norm-attaining functionals annihilating previous spans.

The actual-normal lemma is false

The hoped-for lemma was that finite weighted coordinate energy would force the formal Hilbertian normal to be an element of X . This is false even in the simplest coordinate model. Let $X = c_0$, choose weights $c_n \rightarrow \infty$, and pick a subsequence (n_k) with

$$\sum_k c_{n_k}^{-1} < \infty.$$

Then

$$f = \sum_k c_{n_k}^{-1} e_{n_k}^* \in \ell_1 = X^*$$

satisfies

$$\sum_n c_n |f(e_n)|^2 = \sum_k c_{n_k}^{-1} < \infty.$$

But the formal normal is

$$\sum_n c_n f(e_n) e_n = \sum_k e_{n_k},$$

which does not converge in c_0 . Thus finite coordinate energy alone does not prevent escape into $X^{**} \setminus X$. The extended audit also records a sharper maximizer-level obstruction where two bidual normals cancel.

Literature Status

The exact C^1 source question is still treated as open in the source paper. D’Alessandro–Hajek [2] prove the corresponding positive result for C^k , $k \geq 2$. The supplied audit reports that D’Alessandro’s later treatment and Garcia-Bravo’s 2022 thesis still record the C^1 problem as open. No later full proof or counterexample was located in this run.

The $c_0(\Gamma)$ -injectable case is already answered by Dobrowolski [4, Proposition 5.1]; it should not be advertised as new. The separable proof, quotient-extension result, reflexive-subspace route, and failed-route analysis remain useful for human bookkeeping and for future attempts.

Human Review Checklist

- Verify the Dobrowolski condition (L) identification with bounded injections into $c_0(\Lambda)$.
- Treat Theorems 1 and 2 as correct but mostly known/alternative proofs.
- Check Proposition 1; this is the cleanest new-looking partial extension in this packet.
- Check Proposition 2, especially the infimal-convolution smoothness argument.
- Do not promote this packet to full: the general C^1 question remains open after the recorded push.

References

- [1] D. Azagra, T. Dobrowolski, and M. Garcia-Bravo, *Smooth approximations without critical points of continuous mappings between Banach spaces, and diffeomorphic extractions of sets*, arXiv:1811.07587.
- [2] S. D’Alessandro and P. Hajek, *Polynomial algebras and smooth functions in Banach spaces*, J. Funct. Anal. **266** (2014), 1627–1646.
- [3] R. Deville, G. Godefroy, and V. Zizler, *Smoothness and Renormings in Banach Spaces*, Pitman Monographs and Surveys in Pure and Applied Mathematics, **64**, Longman, Essex, 1993.
- [4] T. Dobrowolski, *Smooth and R -analytic negligibility of subsets and extension of homeomorphism in Banach spaces*, Studia Math. **65** (1979), 115–139.
- [5] S. Dantas, P. Hajek, and T. Russo, *Smooth and polyhedral norms via fundamental biorthogonal systems*, Int. Math. Res. Not. IMRN **2023** (2023), 13909–13939; arXiv:2201.03379.
- [6] P. Hajek and T. Russo, *Norming Markushevich bases: recent results and open problems*, arXiv:2402.18442.

- [7] P. Hajek and J. Talponen, *Some remarks on smooth renormings of Banach spaces*, arXiv:1705.01384.